



MEDIATEK

everyday genius

eCall Overview

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1.0	2023-11-10	Initial release
1.1	2023-03-05	Update API list
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MediaTek Confidential Release for PVT

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1 eCall overview

1.1 eCall introduction

eCall is an in-vehicle road safety system which automatically calls the emergency services in case of a serious accident. As soon as the eCall sensors register a severe impact on a vehicle or a call is initiated manually,

- 1) eCall In-Vehicle System (IVS) establishes e112 voice connection with the relevant Public Safety Answering Point (PSAP).
- 2) Send a Minimum Set of Data (MSD) over the voice connection to PSAP, that includes accurate geo-location data

In April 2015, the European Parliament made it mandatory for all new models of cars to be equipped with eCall technology from 31 March 2018 onward. The eCall system overview is depicted in Figure 2-1.

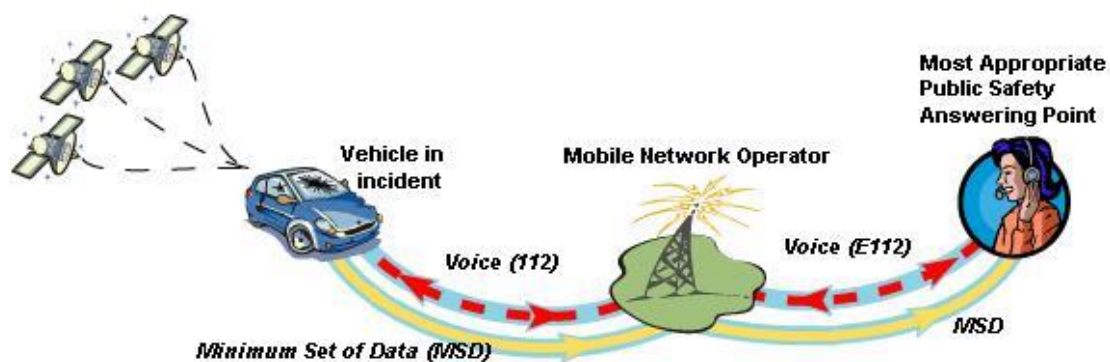


Figure 1-1. eCall system overview

1.2 eCall spec introduction

Basically there are 2 different types of eCall spec, one is for modem protocol such as 3GPP spec, the other is for high layer application spec such EN spec. Generally MTK will cover modem protocol part and provide SDK for customer to integrate high layer application spec with their own application.

Following is a quick glimpse about those different specs. For detail information please refer to the original specs from official websites.

Table 1-1. CS eCall functional spec

Spec name	Description
ETSI TS 126.267 (3GPP TS 26.267)	General description
ETSI TS 126.268 (3GPP TS 26.268)	ANSI-C reference code
ETSI TS 127.007 (3GPP TS 27.007)	6.27 Initiate eCall +CECALL
ETSI TS 122.101 (3GPP TS 22.101)	10.7 Transfer of data during emergency calls
ETSI TS 124.008 (3GPP TS 24.008)	4.4.7 eCall inactivity procedure
3GPP TS31.102	SIM related requirement for eCall
ETSI TS 102 936-1	5. UMTS network access eCall NAD conformance tests 6.GSM network access eCall NAD conformance tests
EN 16062	eCall high level application requirements(HLAP)
EN 16072	PE eCall Operating Requirements
EN 15722	eCall minimum set of data(MSD)
GOST 33465-2015	Protocols of data exchange between in-vehicle emergency call device/system and emergency response system infrastructure
GOST 33464-2015	In-vehicle emergency call device/system. General technical requirement

Table 1-2. CS eCall test spec

Spec name	Description
ETSI TS 126.269 (3GPP TS 26.269)	eCall Data Transfer; In-band modem solution; conformance test
ETSI TS134 123-1	13.3 eCall Emergency Call Procedures
ETSI TS 151 010-1	Conformance specification. 26.9.6a Structured Calls /eCall
ETSI TS103 412	Pan-European eCall end to end and in-band modem conformance testing; Prose test specification
EN16454	Intelligent transport systems –ESafety- Ecall end to end conformance testing
GOST 33467-2015	Functional test methods of in-vehicle emergency call device/system and data transfer protocols
GOST 33470-2015	Test methods for wireless communication modules of in-vehicle emergency call system
ETSI TS 103 428	eCall HLAP Interoperability Testing

Table 1-3. IMS eCall functional spec

Spec name	Description
TS24.229	eCall over IMS(NG-eCall) general spec
RFC8147	NG-eCall SIP part spec
TS23.122	NG-eCall eCall only mode
TS24.301	EMM requirement for eCall only mode
TS23.401	eCall only mode
TS24.008	MM requirement for eCall only mode
TS23.216	NG-eCall SRVCC
TS23.167	eCall over IMS(NG-eCall) general spec
TS27.007	AT CMD of eCall
TS31.102	SIM related requirement for eCall

Table 1-4. IMS eCall test spec

Spec name	Description
TS34.229-1	SIP layer test spec of NG-eCall
TS36.523	UE conformance test spec of NG-eCall

2 eCall SW architecture

The Software architecture is depicted in figure 3-1.

Currently MTK solution supports eCall set-up and in-band modem TX/RX, which conforms to ETSI/3GPP standard (green rectangle), and provides proprietary eCall RIL command.

Custom needs to integrate eCall platform & application (red rectangle) with the eCall RIL command, and uses the product to pass eCall test case.

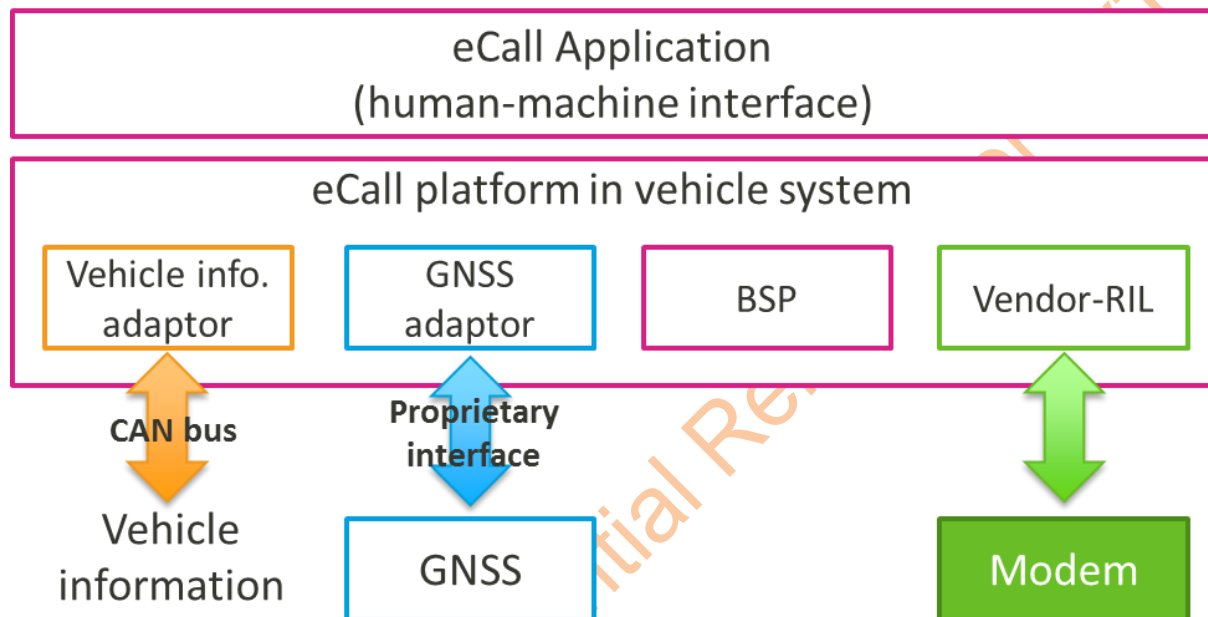


Figure 2-1. eCall SW architecture

3 eCall integration

In this chapter, we introduce the detail flow based on eCall EN standards and how customer can integrate eCall RIL commands to satisfy those standards.

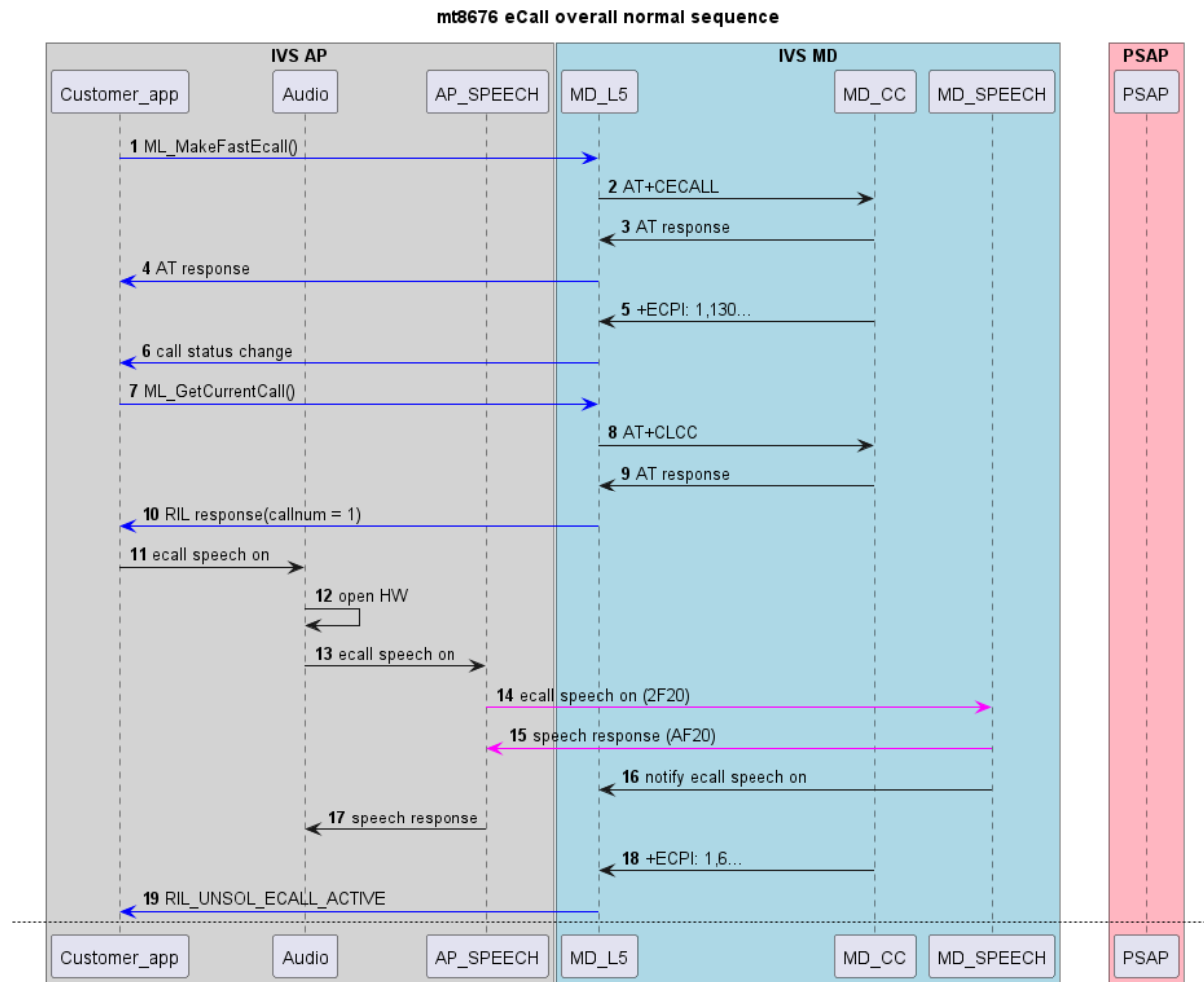
3.1 eCall flow

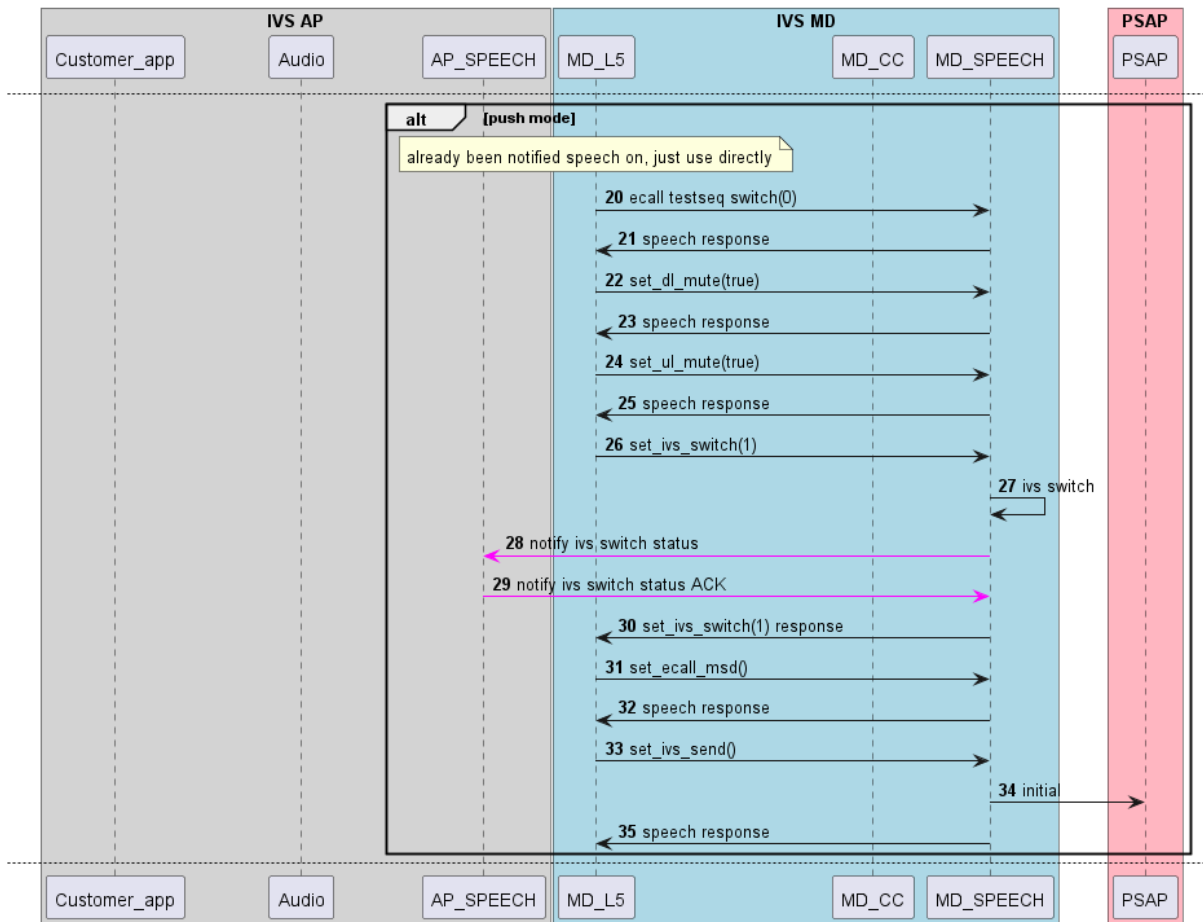
Although there are different eCall scenarios, the keys steps are same. Basically, customer APP need to handle eCall flow in the following steps:

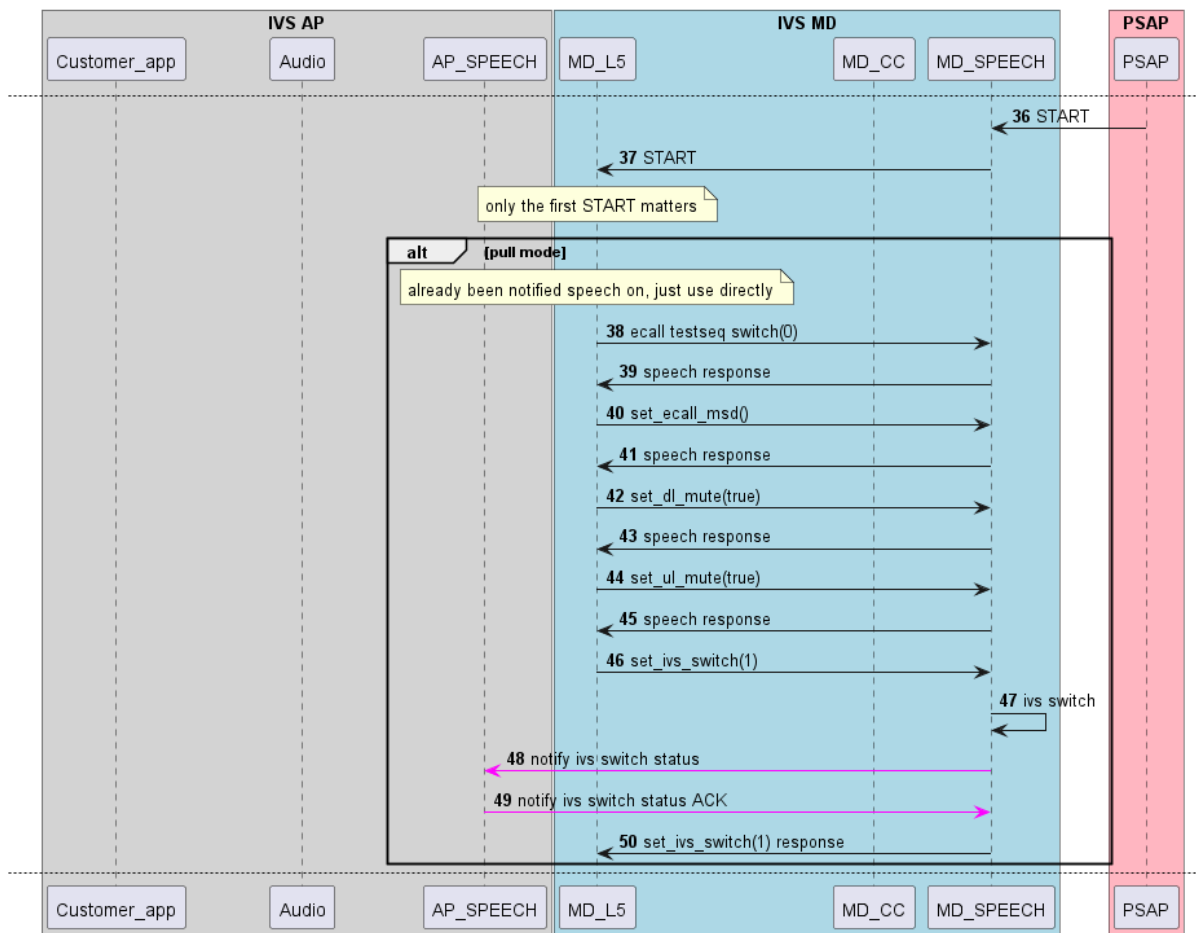
- Initiate an ML_MakeFastEcall
- Maintain eCall timer of IVS side (except T3) which is specified in EN16062.
- Issue RIL_REQUEST_ECALL_RESET_IVS after receiving RIL_UNSOL_ECALL_ALACK_POSITIVE_RECEIVED. Then IVS and PSAP can resume voice call connection and talk to each other.
- Hang up call upon receiving RIL_UNSOL_ECALL_DISCONNECTED or RIL_UNSOL_ECALL_ABNORMAL_HANGUP by which PSAP indicates IVS to disconnect call.

Following is the detailed description of eCall flow sequence

3.1.1 eCall Flow







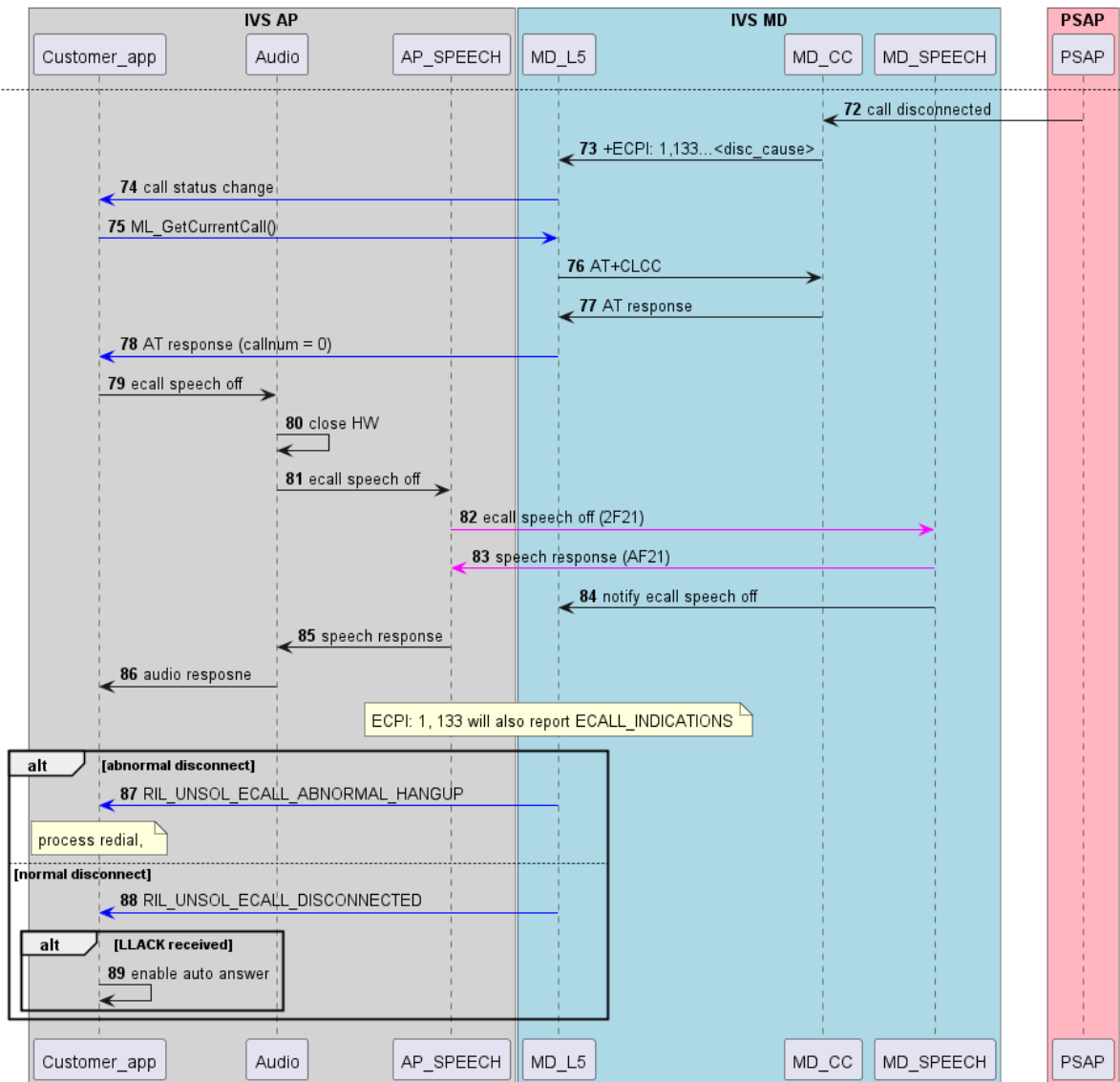


Figure 3-1. eCall flow chart of PSAP

4 Ecall API list

API	function description	Parameter
int32_t ML_ResetIvs(void);	reset eCall state machine in IVS modem, and connect to audio channel	
int32_t ML_SetMSD(ml_ecall_set_msd* msd);	Set MSD data for eCall	typedef struct { int32_t call_id; //call id for ecall uint32_t length; //length of msd_data //Encoded MSD content in hex string, unsigned char msd_data[ML_ECALL_MSD_MAX_LENGTH]; }ml_ecall_set_msd; ex. 34344545, which means 4 values : 52 52 69 69. the length is at most 280 (will be convert to 140 bytes). Input msd is byte type.
int32_t ML_SetTestNumber(ml_ecall_set_num* test_num);	Set test number or URI	typedef struct { //1: only type is valid, 2, type and address is valid int32_t arg_num; //indicate uri type, optional.1 - number, 2 - URI int32_t type; // number or URI, optional char address[128]; }ml_ecall_set_num;
int32_t ML_SetReconfNumber(ml_ecall_set_num* reconf_num);	Set reconfiguration number or URI	typedef struct { //1: only type is valid, 2, type and address is valid int32_t arg_num; //indicate data type, optional.1 - number, 2 - URI int32_t type; // number or URI, optional char address[128]; }ml_ecall_set_num;

API	function description	Parameter
<pre>int32_t ML_MakeFastEcall(ml_ecall_req_ msg* msg);</pre>	Trigger an ecall and transmit MSD data in one command.	<pre>typedef enum { ML_EMER_CAT_MANUAL_ECALL = 1, /* Manual Emergency eCall */ ML_EMER_CAT_AUTO_ECALL = 2, /* Automatic Emergency eCall */ }ml_ecall_category; typedef enum { ML_ECALL_TEST = 1, /* Test eCall */ ML_ECALL_EMERGENCY = 2, /* Emergency eCall */ ML_ECALL_RECONFIG = 3, /* Reconfiguration eCall */ }ml_ecall_variant; typedef enum{ ML_DOMAIN_AUTO = 0, /* Automatic mode - LTE(IMS), WG(CS), 1x(C2K) */ ML_DOMAIN_CS_ONLY = 1, /* CS domain only - WG(CS) */ ML_DOMAIN_3GPP_ONLY = 2, /* 3GPP only - LTE(IMS), WG(CS) */ ML_DOMAIN_3GPP2 = 3, /* 3GPP2 only - 1x(C2K)) */ ML_DOMAIN_IMS_1xCS = 4, /* IMS and 1x CS only - LTE(IMS), 1x(C2K) */ ML_DOMAIN_CS_1x = 5, /* WG CS and 1x CS only - WG(CS), 1x(C2K) */ ML_DOMAIN_IMS_ONLY = 6, /* only IMS call allowed */ }ml_ecall_domain; typedef struct{ ml_ecall_category ecall_cat; ml_ecall_variant ecall_variant; char address[20]; uint32_t length; unsigned char msd_data[ML_ECALL_MSD_MAX_LENGT H]; ml_ecall_domain domain; }ml_ecall_req_msg;</pre>

API	function description	Parameter
int32_t ML_SetEmsdpri(ml_ecall_pri* pri);	set priority of test/reconfiguration eCall number/URI The default priority is “1>3>2>4” .	typedef struct{ int32_t data1; int32_t data2; int32_t data3; int32_t data4; }ml_ecall_pri; The priority is data1>data2>data3>data4, these four parameters should be different from each other and all range from: 1 - Customer assigned eCall URI 2 - eCall URI in USIM 3 - Customer assigned eCall number 4 - eCall number in USIM
int32_t ML_SetNadDeregTime(ml_ecall_time* time);	REQUEST to set nad deregistration time for eCall 设置 T10 timer maintain in md side.	typedef struct{ int32_t purpose; int32_t mode; int32_t timer1; int32_t timer2; }ml_ecall_time; purpose, 0-for ng eCall; 1-for cs ecall mode, 1-set timer; 0-reset timer timer1, timer value (minute) set for emergency call timer2, timer value (minute) set for rest/reconfiguration call, in current, timer1 and timer2 prefer to be the same value.
int32_t ML_SetOprrMode(ml_ecall_oprr_mode mode);	Request to set the service status for operating mode.	typedef enum { ECALL_NONE = 0, ECALL_ONLY = 1, ECALL_AND_NORMAL = 2, }ml_ecall_oprr_mode;

API	function description	Parameter
int32_t ML_GetOprtMode(ml_ecall_oprt_mode mode);	Request to get the service status for operating mode.	
int32_t ML_EcallIndicationInit(ML_ECALL_MSGCB_T cb_func);	Register ecall indication call back. App will receive ecall indication after init ecall indication.	typedef enum{ E_ML_ECALL_SENDING_START = 1, E_ML_ECALL_SENDING_MSD = 2, E_ML_ECALL_LACK_RECEIVED = 3, E_ML_ECALL_ALACK_POSITIVE_RECEIVED = 4, E_ML_ECALL_ALACK_CLEARDOWN_RECEIVED = 5, E_ML_ECALL_DIALING = 9, E_ML_ECALL_ALERTING = 10, E_ML_ECALL_ACTIVE = 11, E_ML_ECALL_DISCONNECTED = 12, E_ML_ECALL_IMS_ACTIVE = 13, E_ML_ECALL_IMS_DISCONNECTED = 14, E_ML_ECALL_ABNORMAL_HANGUP = 15, E_ML_ECALL_IMS_MSD_ACK = 20, E_ML_ECALL_IMS_UPDATE_MSD = 21, E_ML_ECALL_IMS_IN_BAND_TRANSFER = 22, E_ML_ECALL_IMS_MSD_NACK = 23, E_ML_ECALL_IMS_SRVCC = 24, E_ML_ECALL_ONLY_DEREGISTRATION = 31, E_ML_ECALL_MAY_DEREGISTER = 32, E_ML_ECALL_PSAP_CALLBACK_START = 40, E_ML_ECALL_PSAP_CALLBACK_IMS_UPDATE_MSD = 41, E_ML_ECALL_T2_TIMEOUT = 52, E_ML_ECALL_T5_TIMEOUT = 55, E_ML_ECALL_T6_TIMEOUT = 56, E_ML_ECALL_T7_TIMEOUT = 57, E_ML_ECALL_UNSPECIFIED = 0xffff, }ML_ECall_Indication;

API	function description	Parameter
		<pre> typedef struct { ML_ECall_Indication ind; int call_id; } ML_ECALL_IND_T; typedef void (*ML_ECALL_MSGCB_T) (ML_ECALL_IND_T *pvsMsg); </pre>

5 eCall Timer

5.1 EU CS eCall timer

In an eCall transaction, there are 10 timers defined in EN 16062. MTK has implemented T3 and T10 in modem side. T1,T2,T5,T6,T7,T9 should be handled by IVS customer APP; T4 and T8 are PSAP timers and should be handled by PSAP application. In order to pass instrument test, mt2731 demoapp has reference code for T2/T5/T6/T7.

Table 4-1. eCall timers in EN16062

Name Origin	Description Requirements	Value
T1 IVS	Manually initiated eCall(MIeC) false triggering cancellation period - START: T1 starts as soon as the eCall is manually activated - STOP: T1 stops when Vehicle occupants cancel the manually triggered eCall transaction. - EXPIRY: Upon expiry of T1 the IVS-NAD shall start call setup	See NOTE1 See NOTE2
T2 IVS	IVS Call Cleardown Fallback Timer (CCFT) - START: T2 starts as soon as the IVS-NAD starts with call setup - STOP: T2 stops when the IVS-NAD receives a call clear-down indication from the mobile network or a call clear-down message from the PSAP. - EXPIRY: Upon expiry of T2 the IVS-NAD shall clear down the call	3600 s (1h)
T3 IVS	IVS INITIATION signal duration - START: T3 is started as soon as the IVS-NAD starts sending the INITIATION signal - STOP: T3 stops when the IVS-NAD receives a SEND MSD signal from the PSAP, at which time the IVS-NAD shall stop sending the INITIATION signal. - EXPIRY: Upon expiry of T3 the IVS-NAD shall stop sending the INITIATION signal	2 s
T4 PSAP	PSAP wait for INITIATION signal period	5 s

	• START: T4 starts as soon as the PSAP eCall modem has answered the call • STOP: T4 stops when the PSAP eCall modem detects an INITIATION signal send by the IVS. • EXPIRY: Upon expiry of T4, the PSAP eCall modem shall route the call to a PSAP operator	
T5 IVS	IVS wait for SEND MSD period • START: T5 starts as soon as the IVS-NAD received notification that the call is first answered • STOP: T5 stops when the IVS-NAD detects a SEND MSD signal sent by the PSAP. • EXPIRY: Upon expiry of T5 the IVS-NAD shall reconnect the IVS audio system and terminate eCall specific behavior(i.e. it shall not proceed with the sending of MSD data) until requested to do otherwise.	5 s
T6 IVS	IVS wait for AL-ACK period • START: T6 starts as soon as the IVS-NAD has received LL-ACK • STOP: T6 stops when the IVS-NAD receives an AL-ACK message • EXPIRY: Upon expiry of T6, the IVS-NAD shall mark the transfer of the MSD as unsuccessful and reconnect the IVS audio system and terminate eCall specific behavior until requested to do otherwise	5 s
T7 IVS	IVS MSD maximum transmission time • START: T7 starts as soon as the IVS-NAD starts sending the MSD data • STOP: T7 stops when the IVS-NAD receives an LL-ACK message • EXPIRY: Upon expiry of T7, the IVS-NAD shall mark the transfer of the MSD as unsuccessful and reconnect the IVS audio system and terminate eCall specific behavior until requested to do otherwise	20 s
T8 PSAP	PSAP MSD maximum reception time	20 s

	• START: T8 starts as soon as the PSAP starts sending the SEND MSD signal • STOP: T8 stops when the PSAP eCall modem receives a valid MSD (reception being acknowledged by sending an LL-ACK) • EXPIRY: Upon expiry of T8, the PSAP eCall modem shall route the call to a PSAP operator	
T9 IVS	IVS NAD minimum network registration period • START: T9 starts as soon as the IVS-NAD clears down a call, or gets notified that a call has been cleared down in accordance with EN 16072 Clause 7.17.3 • STOP: T9 is uninterruptable; until T9 expires the IVS-NAD shall remain registered on the serving network, and remain available to receive calls from the PSAP and rescue workers • EXPIRY: Upon expiry of T9, the IVS-NAD may deregister from the serving network (see T10)	3600 s
T10 IVS	IVS NAD network ‘Deregistration Fallback Timer’ (DFT) • START: T10 starts as soon as the IVS-NAD clears down a call, or gets notified that a call has been cleared down • STOP: T10 stops if the IVS-NAD receives or makes a new call • EXPIRY: Upon expiry of T10, the IVS-NAD shall deregister itself from the serving network	12 h see no

NOTE 1 No normative value, can be chosen by manufacturer, zero(“0”, effectively disabling T1) is allowed.

NOTE 2 T1 is not relevant for automatically activated eCall transactions.

NOTE 3 only for ‘eCall only’ IVS devices, timer not relevant for IVS NAD devices that remain connected to a network.